# Silicon Nanowires and Lithium Cobalt Oxide Nanowires in Graphene Nanoribbon Papers for Full Lithium Ion Battery

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**Described here is the production and characterization of a scalable method to produce 3D structured lithium ion battery anodes using free-standing papers of porous silicon nanowires (Si-NW) and graphene nanoribbons (GNRs). Using simple filtration methods, GNRs and Si-NWs can be entangled into a mat thereby forming Si-NW GNR papers. This produces anodes with high gravimetric capacity (up to 2500 mA h $g^{-1}$) and high areal and volumetric capacities (up to 11 mA h $cm^{-2}$ and 3960 mA h $cm^{-3}$). The compact structure of the anode is possible since the GNR volume occupies a high proportion of empty space within the composite paper. These Si-NW/GNR papers have been cycled for over 300 cycles, exhibiting a stable life cycle. Combined with $LiCoO_2$ nanowires, a full battery is produced with high energy density (386 Wh $kg^{-1}$), meeting requirements for high performance devices.**

## 1. Introduction

The widespread use of portable electronic devices has increased the demand for high performance energy storage systems.[1] Present and future applications such as hybrid and total electrical vehicles, electric tools, and portable devices, such as smartphones, require lithium ion batteries (LIBs) with improved features. These features include higher capacity and power, lower net weight, and extended cycle-life. For different applications, the materials need to be varied to maximize performances.[2–4] Currently, the most widely used battery technology is based on an anode composed of graphite, which has a theoretical capacity of 372 mA h $g^{-1}$, and a cathode based on lithiated metal oxides, such as $LiCoO_2$ (LCO), whose total capacity is generally below 150 mA h $g^{-1}$. These two materials compose LIBs with a high voltage operation of ≈3.7 V, and relatively high energies densities of 100–250 Wh $kg^{-1}$, but with limited capacity of ≈100 mA h $g^{-1}$. Therefore new anodes and cathodes with higher specific capacities are sought.

After several years of research, the capacities of high voltage metal oxide cathodes remained below 200 mA h $g^{-1}$, although a few examples have attained small increments of capacity or higher voltage operation >5 V versus $Li/Li^+$.[5–7] An exception is sulfur, which has an outstanding capacity of 1675 mA h $g^{-1}$, but a lower voltage of operation of 2.1 V.[8] On the other hand, several high capacity alternatives, such as Li metal, Si, and other alloying metals (Sn, Sb, Al) are possible in anode applications.[9] This higher capacity can decrease the total amount of anode mass necessary to compose the batteries, increasing the energy density. In this sense, Si is particularly attractive due to its practical capacity of ≈3500 mA h $g^{-1}$ (the theoretical capacity is 3579 mA h $g^{-1}$ at room temperature, based on $Li_{15}Si_4$ alloy),[10] its earth abundance, environmental friendliness, stability at ambient atmosphere and its average delithiation potential of ≈0.4 V, which is similar to graphite's of 0.15 V versus $Li/Li^+$. Hence, Si is compatible with the same high voltage operation of ≈3.7 V when combined with commercial cathodes.[11]

The hurdles associated with Si anodes, such as its large volume expansion of up to 300%, low electronic conductivity and high reactivity with common electrolytes, can compromise its stability and capacity.[12] This has been previously addressed by employing capped Si nanostructures to prevent pulverization,[13,14] by using Si/C nanocomposites,[15] by limiting the extension of Li reaction with Si[16] and by using specifically designed electrolytes to control the solid electrolyte interphase (SEI) formation.[17,18] The SEI should be stable and it is especially important in materials that develop high volume expansion during Li uptake, or else the cracking can result in electrolyte decomposition over the bare Si surface. These strategies have generated anodes that deliver gravimetric capacities much higher than those seen in graphite. However, a higher gravimetric or specific capacity is by no means the most important factor to outperform graphite as an anode in LIBs.[19,20] A material with a higher specific capacity must also be distributed over a small area because of the limited size of LIB electrodes, displaying a competitive areal and volumetric capacity that at least meets current industry standards for graphite, which is 2–4 mA h $cm^{-2}$ and >600 Wh $L^{-1}$. Most Si anode half-cells have been reported to have high specific capacities of 1000–3000 mA h $g^{-1}$, but there is limited information regarding their areal capacity.[20] Noting that cycle life and electrode thickness are inversely proportional,[21,22] a reliable comparison among electrodes becomes exceedingly difficult without fully understanding the gravimetric, areal, and volumetric capacities.

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Hierarchical structures such carbon buckypapers made from carbon nanotubes or graphene analogue papers, have been used in capacitors and batteries due to their ease of preparation, high electronic conductivity, and their bulk 3D and free-standing materials properties.[23] In this work, we report a simple filtration-type "paper" composed of silicon nanowires (Si-NWs) and graphene nanoribbons (GNRs), that provides a compact Si electrode for LIBs. These materials show high gravimetric capacity of 2000–2500 mA h $g^{-1}$ with excellent areal and volumetric capacities of ≈11 mA h $cm^{-2}$ and 3960 mA h $cm^{-3}$, respectively. The conductive and compact structure of the electrodes can be supported by GNRs alone as the current collector. The remainder of the structure is composed of a mixture of GNRs for improving the conductivity and Si-NWs for the Li storage. The resulting Si-NW/GNR paper produced by co-filtration of the two materials enables not only an open structure accessible to the electrolyte, but also multiple electrical connections between the GNRs and Si-NWs due to the duality of the two high aspect ratio materials that are resilient during the continuous cycling. Combined with a $LiCoO_2$ nanowire (LCO-NW) cathode, a full battery is demonstrated with a high gravimetric energy density of 386 Wh $kg^{-1}$ and an average potential of 3.65 V, similar to that seen in commercial graphite-based LIBs.

## 2. Results and Discussion

**Figure 1**a displays the scheme for preparing compact free-standing paper made from Si-NWs and GNRs. The porous Si-NWs were produced by chemical etching from highly

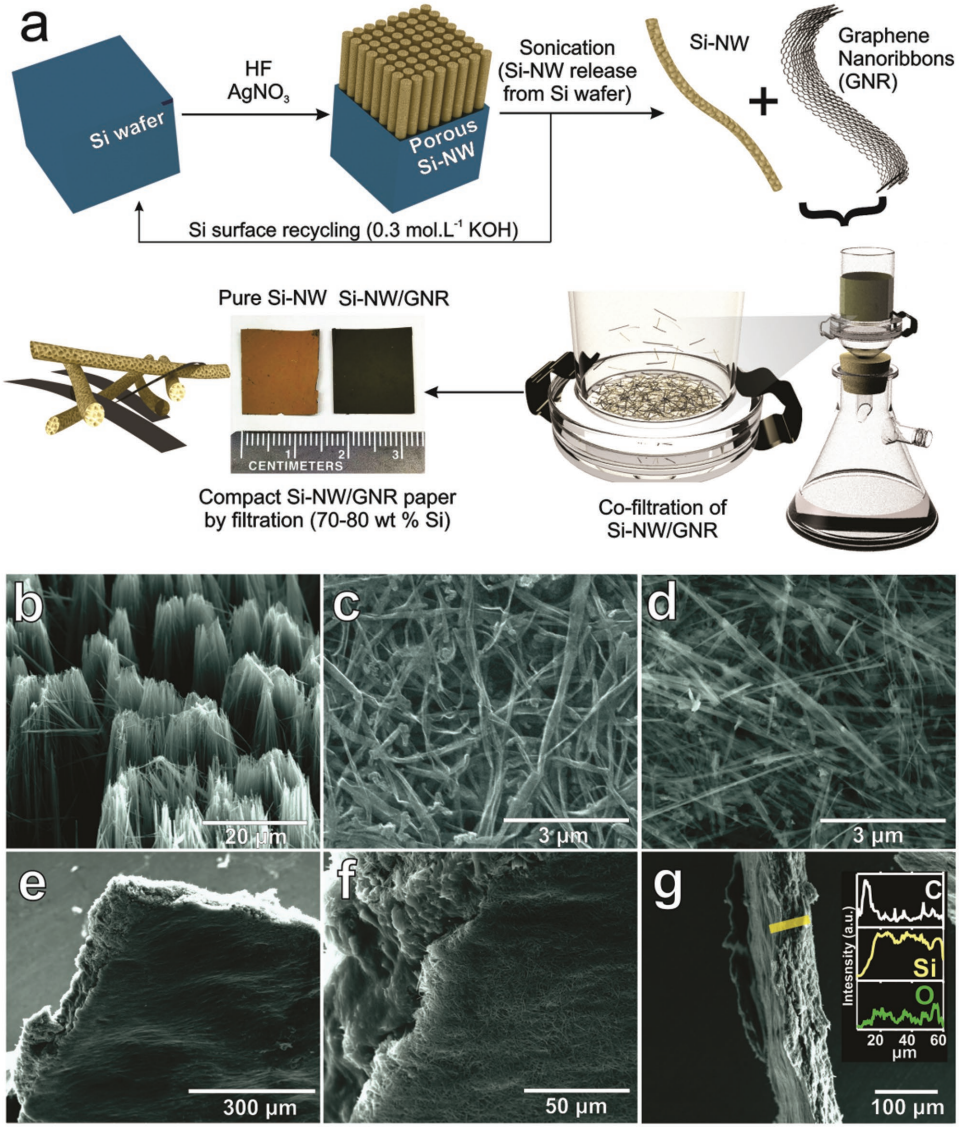



**Figure 1.** a) Scheme for the preparation of porous Si-NW/GNR papers. b) SEM image of the Si-NW forest on the Si wafer before Si-NW removal. c) SEM image of the pure GNR paper. d) SEM image (top-view) of the hybrid Si-NW/GNR paper. e,f) SEM images of bottom-view of Si-NW/GNR paper. g) The cross-section SEM of the Si-NW/GNR paper. Inset: Line scan mapping by EDAX of C, Si, and O across the film. The yellow line of the line scan is 60 µm in length, while the inset figure bear a scale bar accordingly to their own *x*-axis.


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boron-doped Si wafers in a one step preparation (see the Experimental Section).[24] The metal-assisted chemical etching method is known to be a versatile protocol to produce porous Si-NWs from different Si sources such as Si wafers (with different degrees of *p* or *n*-doping) or from preactivated bulk Si particles.[25,26] The produced Si-NWs could be easily removed from the surface of wafers by employing a low power sonicator (see the Experimental Section), generating isolated Si-NWs that have a broad distribution of diameter of 10–100 nm and average lengths of 10 µm. The porous structure are the result of the high doping level which enables more active sites for the etching solution. After removal of the Si-NWs, the wafer can be recycled to prepare more Si-NWs. This was done by etching the root part of the wires that could not be removed by sonication using a KOH solution (0.3 mol $L^{-1}$) and then following the same process to generate new Si-NWs. The scanning electron microscopy (SEM) images of all materials after each step are presented in Figure S1 in the Supporting Information and Figure 1b–g. The top-view SEM images of the vertically aligned forest of Si-NWs are displayed in Figure 1b. The isolated Si-NWs are then mixed with GNRs in solution and co-filtered to compose the active layer of the electrode (Figure 1a), in which the Si represents 70%–80% of the mass of the electrode. The GNR preparation was previously described, and is based on reductive splitting of multiwalled carbon nanotubes using liquid NaK alloy (Figure 1c).[27] Raman spectra and X-ray diffractograms of the materials and the Si-NW/GNR films are presented in Figure S2 in the Supporting Information. Evidence of the cubic (diamond-like) crystalline structure of the Si and the $sp^2$ signature of graphene are seen. Photographs of the resulting free-standing papers are presented in Figure 1a, in which a darker color is observed in the mixture of GNRs and Si-NWs. Si-NWs alone have a yellow color. The GNRs also improve the mechanical properties of the Si-NW films making the nanocomposite paper more robust and flexible when compared to the more brittle pure Si-NW papers. The high aspect ratio of both materials is a helpful feature of this composition since it allows the Si-NWs and GNRs to have multiple contact points forming a stable conductive path across the electrode. Also, the GNRs can percolate through the empty spaces between the Si-NWs and they do not contribute significantly to the volume of the electrode. An SEM image (top view) of the compact Si-NW/GNR anode is shown in Figure 1d, showing the entangled structure of Si-NWs and GNRs that work as a mat-like paper electrode.

A pure GNR dispersion can also be filtered first to form a bottom layer electrode made of a thin pure GNR film that can work as current collector, avoiding the use of metal, such as Cu foil. This lowers the total mass of the anode. Figure 1e,f shows the bottom part of the Si-NW/GNR films composed solely of GNRs. The high concentration of GNRs at the bottom and the homogeneity of GNRs across the depth of the electrode was checked by imaging the cross-section of the Si-NW/GNR film (Figure 1g) and analyzing the Si, O, and C signals by energy dispersive X-ray spectroscopy (EDAX) (inset of Figure 1g). The line scan analysis, corresponding to the dotted yellow line in Figure 1g, confirms the homogeneity of Si and O from native surface $SiO_x$ species. As expected, the C signal is more intense only at the bottom of the electrode, where the Si content drops to near zero.

The porous nature of the Si-NW generates empty spaces within the crystalline structure of the Si-NW, which can lower the stress caused by the high volume expansion during Li uptake.[28] Transmission electron microscopy (TEM) of the Si-NW, GNR, and mixtures of Si-NW and GNR are presented in **Figure 2**. The porous Si-NW images (Figure 2b–e) show that a variety of sizes and even bundles of Si-NWs can be

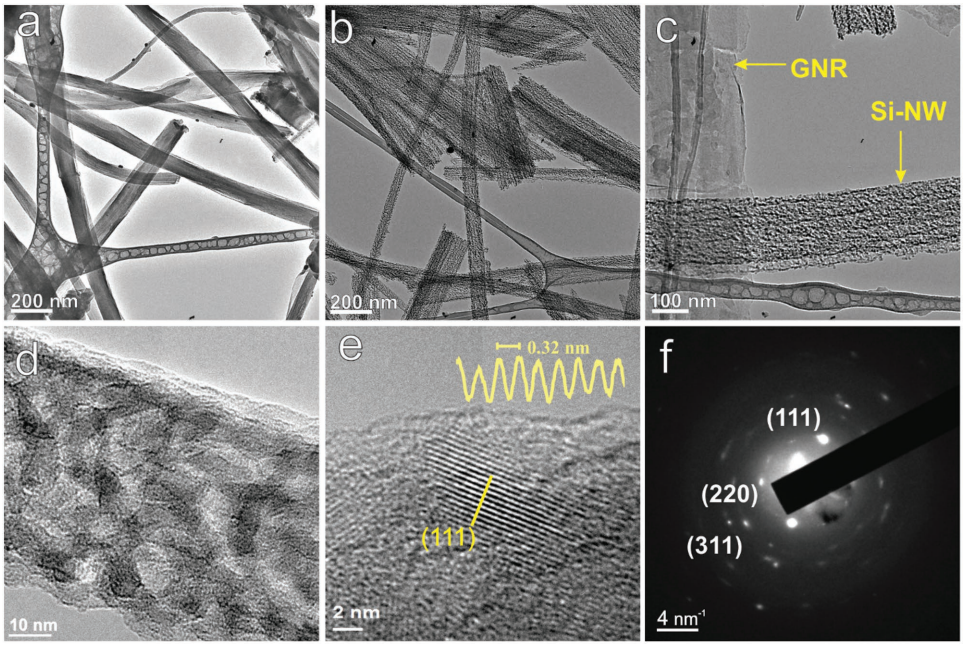



**Figure 2.** a) TEM images of GNRs on a carbon grid. b) TEM images of Si-NWs on a carbon grid. c) TEM image of the contact of a GNR and Si-NW. d,e) High magnification TEM images of the porous structure of Si-NW. Inset: spacing of (111) planes of Si. f) SAED of a Si-NW and the corresponding planes.

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found, and the porous nature is seen by the contrast of the nanowires (Figure 2b). Despite the different diameters of the Si-NWs, small (<10 nm) crystalline domains are observed distributed homogenously in the entire volume of the nanowire (Figure 2d). These crystallite sizes are well-below the critical level of 150 nm, the minimal estimated size to avoid intense fracture on the surface of the particles (pulverization process) during the delithiation reaction.[12,29] The high resolution TEM image (Figure 2e) shows the diamond-like structure of the Si-NW as the corresponding (111) atomic planes (0.32 nm) can be identified (inset of Figure 2e, contrast profile of the atomic planes (111)). Evidence of the $SiO_x$ layer on the surface of the wires is also observed in Figure 2d,e. Selected area electron diffraction (SAED) shows that the nanowires are single-crystalline (Figure 2f). The mixture of Si-NW and GNR was also inspected (Figure 2c), as to compare to images of pure GNRs in Figure 2a. The simple mixture shows that several contact points of entangled Si-NWs and GNRs are obtained, with the flat structure of the GNRs touching several Si-NWs as expected for two materials with high aspect ratios. This further mitigates Si-NW bundling inside the electrode; had bundling occurred, the overall resistance of the anode would have increased. As in the case of random nanowires networks, higher aspect ratio materials enable more contact points, minimizing the overall resistance of the paper.[30] Current versus potential measurements tested across the electrode shows good conductivity throughout the extension of the electrode (**Figure 3**). We observed that the conductivity was highly dependent on the compactness of the Si-NW/GNR film. During coin cell sealing, a pressure of 1000 psi (see the Experimental Section) was applied to the anode film reducing its thickness and increasing the compactness (Figure S3, Supporting Information). The measurements performed on the as-produced films (no pressing) showed much smaller conductivities with an average value of 560 S $m^{-1}$ (Figure S3, Supporting Information), much higher than similar Si-NW films reported elsewhere.[31] In the Si-NW/GNR films after pressing (curves in Figure 3), a much higher conductivity was measured with an average value of 2568 S $m^{-1}$ and a maximum conductivity 3597.2 S $m^{-1}$ (curves in Figure 3).

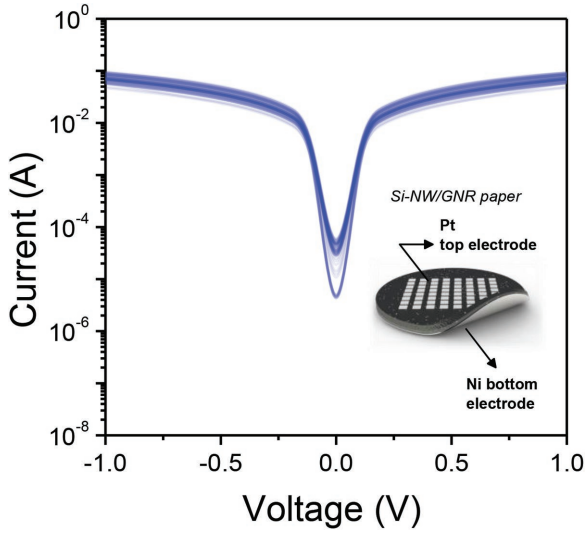



**Figure 3.** Current–potential curves of the Si-NW/GNR paper measured across the film after pressing. Inset: scheme of measurements using metal layers (Pt—top and Ni—bottom) evaporated on the top and bottom surface of the paper. Curves correspond to several tests performed in different spots on the top of the electrode.

### 2.1. Electrochemical Characterization

The conductivity data on the entangled structure of Si-NW/GNR papers show that the conductivity is nearly constant regardless of thickness (Figure 3; Figure S3, Supporting Information). Galvanostatic charge–discharge curves of the anode are shown in **Figure 4** for determining the total capacity for lithiation in terms of mass, area, and volume of the electrode in the range of 0.01–1.3 V. The first cycle of discharge (Figure 4a) (lithiation reaction, formation of $Li_xSi_y$ alloy) displays a large irreversible capacity, as it known for Si anodes, tested at 0.2 A $g^{-1}$, resulting in an overall low coulombic efficiency in the first cycle (48%). The capacity loss in the first cycle is related to the formation of SEI and can be largely reduced depending on the electrolyte composition and coatings over the Si anode structures.[14,16,17] In our case, the significant capacity loss requires the use of prelithiation methods when considering the capacity balance to produce the full battery, as will be discussed. The subsequent cycles show reversible capacity of 2000–2500 mA h $g^{-1}$, highly superior to the specific capacity of graphite (indicated as dashed line in Figure 4a). The alloying reaction in the first cycle follows the expected process at 0.1 V for lithiation of crystalline Si. After the first process of lithiation, the formation of $Li_xSi_y$ starts below 0.3 V, typical of amorphous Si.[32,33] Both processes can be clearly observed in the cyclic voltammograms (CVs) (Figure 4b), as well as the delithiation reactions. Si-NW/GNR anodes with different Si loadings per area of electrode were also tested, resulting in different areal capacities from ≈1.5 to 11 mA h $cm^{-2}$, comparatively higher than commercial requirements of 2–4 mA h $cm^{-2}$ (dashed line in Figure 4c positioned at 4 mA h $cm^{-2}$).[20] Despite the high capacity accumulated per area of electrode, the voltage profile (polarization between charge and discharge) does not change significantly with the increment of Si-NW per area. Regardless of the difference of thickness in these electrodes, the compact but open structure provided by the entangled Si-NW/GNR electrode permits homogenous access of the electrolyte to the surface of the Si-NWs. The highly compact structure of the anodes also leads to a high volumetric capacity of ≈3980 mA h $cm^{-3}$ (Figure 4d), one of the highest present in the literature and close to the theoretical volumetric capacity expected for packed Si particles.[14,19] Recently, Peled et al. reported a high areal capacity Si anode using Si-NWs grown inside carbon fiber paper.[20] Although high capacity per area was achieved (up to 22 mA h $cm^{-2}$), the Si-NW are located within the volume of the carbon fiber and therefore the volumetric capacity is below 700 mA h $cm^{-3}$. In our case, we can keep the same areal capacity but considerably higher volumetric capacity.

The pure Si-NW paper performance has lower volumetric capacitance due its lower conductivity, while the pure GNR anode yields a volumetric capacity of 450 mA h $cm^{-3}$, comparable to the value found in graphite anodes (550 mA h $cm^{-3}$).


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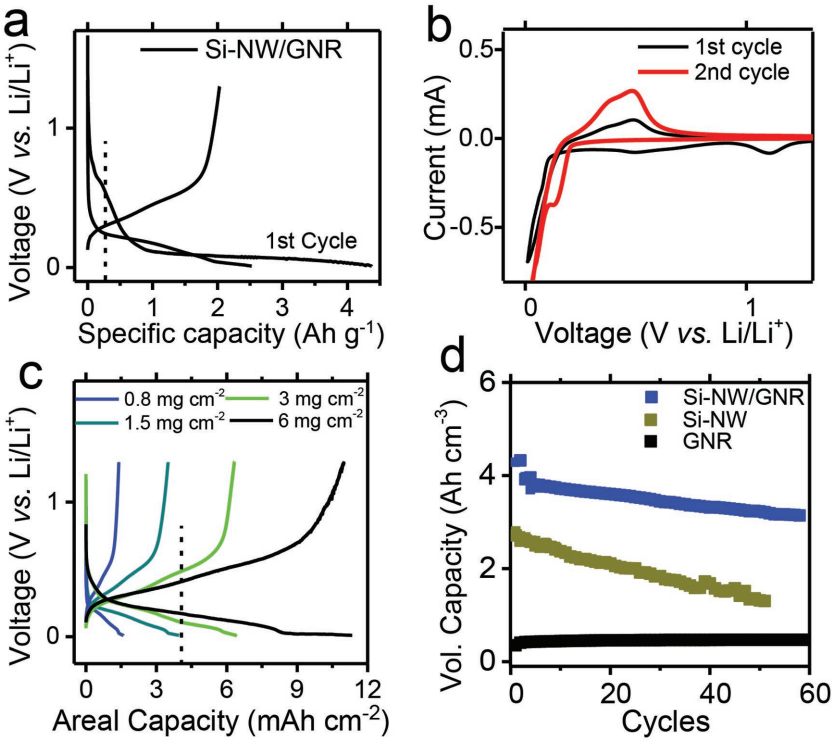



**Figure 4.** a) Charge–discharge profile of Si-NW/GNR paper with the first two cycles presented at 0.2 A $g^{-1}$. b) CVs of Si-NW/GNR paper at 0.1 mV $s^{-1}$. c) Areal capacity control of Si-NW/GNR paper through mass density per area of electrode. d) Volumetric capacity stability through cycling (0.2 A $g^{-1}$).

Evidently, GNRs are also active in the lithiation reaction; however, its contribution to the total capacity of the Si-NW/GNR paper is minimal and less than 5% in terms of total capacity. This was calculated by applying the same current density of a Si-NW/GNR half-cell to a half-cell battery containing only GNRs. While the applied current density is equivalent to 0.2 A $g^{-1}$ to the Si mass, it is equivalent to 0.8 A $g^{-1}$ for pure GNRs, leading to a lower capacity and therefore lower contribution in the Si-NW/GNR film capacity.

The high volumetric capacity can be justified by the lower volume contribution of GNRs throughout the Si-NW/GNR compact paper electrode. The cross-sectional image of the GNR films, Si-NW films and different mass loadings of Si-NW/GNR are shown in Figure S4 in the Supporting Information (images taken after the coin cell assembly for at least five cycles). Si-NW papers of 6 mg $cm^{-2}$ with and without GNRs present similar thicknesses of ≈40 µm (Figure S4, Supporting Information). Apparently, there is a large void space among the randomly packed Si-NWs. In fact, mathematical models show that low volume fractions (total volume occupied by the constituents particles) of less than 30% are found in materials with high aspect ratio where length/diameter ($L/D$) > 30.[34] An estimate of $L/D$ of the Si-NW would be ≈100 (for GNR, $L/D$≈10, although the relationship is valid for rod-like structures, not ribbons), considering the average values of diameter and length. The empty space is occupied by the GNR, percolating and generating a stable and conductive interpenetrated matrix inside the Si-NW packing. The network of GNRs should remain stable during the lithiation/delithiation reaction.

In **Figure 5**, we tested the rate performance of Si-NW/GNR anodes and the cycling rate performance of the electrodes from 0.3 to 10 A $g^{-1}$. We observe high capacities at high rate with little effect on the polarization up to 1 A $g^{-1}$. At higher rates, the measured polarization was high (0.5–0.7 V) (Figure 5a); however, the capacity tested at these higher rates, up to 5 A $g^{-1}$, still outperforms the values found in graphite at lower rate (<0.1 A $g^{-1}$) (Figure 5b).[35] The capacity response at different rates is dependent on the Li diffusion along the thickness direction, which in turn depends on the conductivity, SEI composition, and crystalline structure of the material.[36] After testing under high rate, the capacity can be recovered if tested

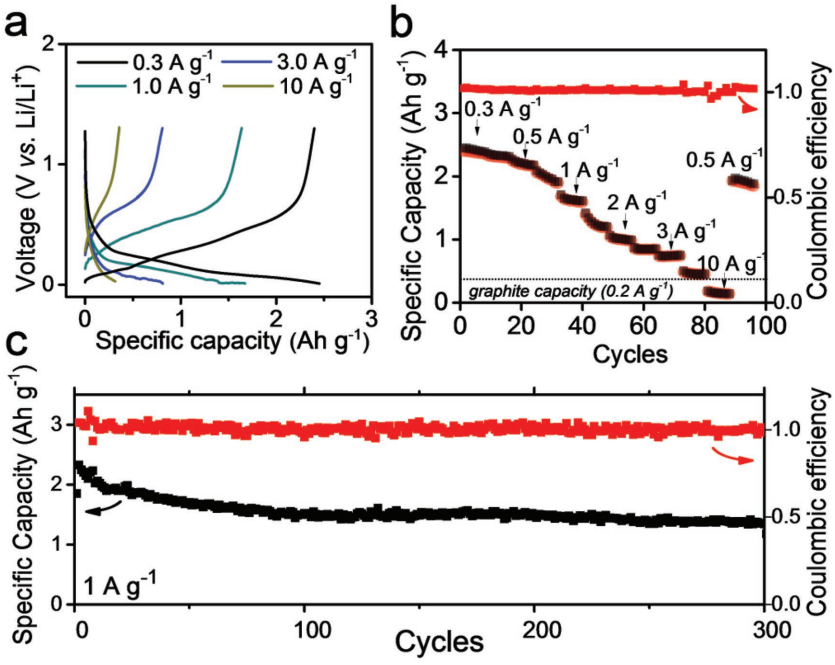



**Figure 5.** a) Charge–discharge profile of Si-NW/GNR at different rates. b) Cycling stability at different rates. c) Long term cycling stability at 1 A $g^{-1}$.


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again under lower rates (0.5 A $g^{-1}$) (Figure 5b). The long-range stability is also demonstrated at 1 A $g^{-1}$ (Figure 5c). Despite a loss at the initial cycles in this test, a high capacity is observed (≈1500 mA h $g^{-1}$) with a coulombic efficiency above 99.6%.

In order to compose a full cell, we also performed studies using LCO-NWs synthesized by using an adapted hydrothermal method (see the Experimental Section),[37] and combining them with GNRs to produce LCO-NW/GNR papers by filtration. The LCO-NWs with their characterization data are shown in Figure S5 in the Supporting Information. The half-cell tests with the galvanostatic charge–discharge curves are shown in Figure S6 in the Supporting Information. Similar high areal capacities can be achieved per electrode, up to 5 mA h $cm^{-2}$. The specific capacities at 0.14 A $g^{-1}$ for LCO-NW was ≈150 mA h $g^{-1}$, close to the best values of $LiCoO_2$ reported.[37,38] **Figure 6**a shows the voltage profile (versus $Li/Li^+$) of both Si-NW/GNR anodes and LCO-NW/GNR cathodes. The high difference in terms of specific capacity required an imbalance in terms of the mass of the cathode and anode, proportional to the ratio of their capacities ($Q_{Si}$≈12 $Q_{LCO}$). The full battery curves (Figure 6b) are the result of the curves observed in the half-cells: the difference of the full lines in Figure 6a (discharge of Si and charge of $LiCoO_2$) gives the charge curves in Figure 6b, while the dotted curves (charge in Si and discharge of $LiCoO_2$) resulted in the discharge of the full cell (tested with a rate of 0.2 A $g^{-1}$). The difference of these curves also gives us the safe voltage limits of charge at 4.05 V and discharge at 2.9 V. The flat voltage interval ranges from 4.0 to 3.3 V, with an intermediate value of 3.65 V, close to the values normally found in the graphite/$LiCoO_2$ systems at 3.7 V. This voltage operation during the discharge is enough to power electronic devices that generally require this higher voltage, such as the LCD screen of smartphones, as presented in the inset of Figure 6a, in which a commercial LCD screen is powered by a coin cell with a total capacity of 2 mA h. As the capacity of the cathode is much lower than the anode, the specific capacity of the full cell is near the specific capacity of the pure cathode (top axis, Figure 6b), close to 110 mA h $g^{-1}$.

We also observed that the performance of the full batteries were greatly compromised if a fresh Si anode was directly combined with a fresh $LiCoO_2$, which happens because of the large irreversible capacity in the first cycle of the Si-NW/GNR anodes (first cycle coulombic efficiency ≈48%, as discussed previously), leading to the irreversible consumption of Li ions from $LiCoO_2$. The irreversible capacity of the first cycle is an important issue in commercial LIBs.[38] Because of this inherent capacity loss associated with the first cycle, the Si-NW/GNR anodes were first cycled in half-cells or prelithiated by applying an Li foil wetted with electrolyte over the fresh anode for 2 h (see the Experimental Section).[39,40] These two methods enabled low irreversible capacity after the first charge cycle of the full battery (Figure 6b), since the SEI formation was previously achieved at the expense of the external Li source (Li foil) rather than the LCO-NW/GNR, thus preserving the cathode capacity. Therefore, the mass balancing between anode and cathode ($m_{LCO}$≈12 $m_{Si}$) considered the capacity balancing based on the capacity ratio $Q_{Si}$≈12 $Q_{LCO}$, as found in the half-cells measurements. We also tested the self-discharge property of the full-cells (Figure 6c). The self-discharge measures the stability of the lithiated state of the Si in the full battery without an applied external voltage.[41] After full charge, the full battery exhibited a good retention of both voltage and capacity, ≈90% (Figure S7, Supporting Information) after 120 h.

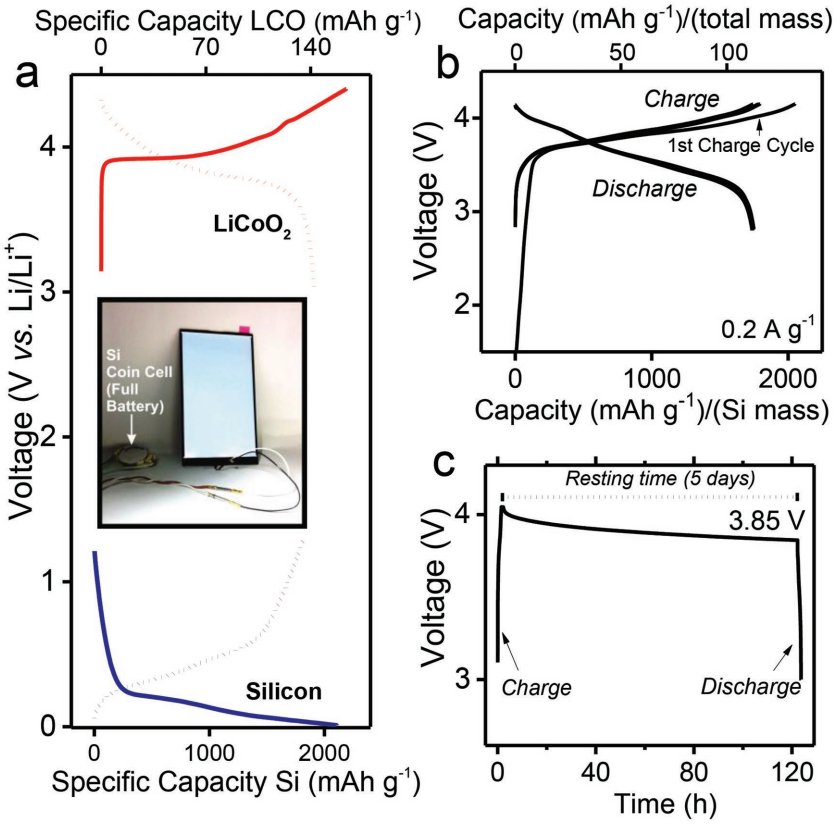



**Figure 6.** a) Charge–discharge profile of Si-NW/GNR and LCO-NW/GNR half-cells. Inset: Full battery powering a smartphone LCD screen. b) Charge–discharge profile of the full battery Si-NW/GNR and LCO-NW/GNR (4.05–2.9 V) at 0.2 A $g^{-1}$. c) Self-discharge test for 120 h after full charge to 4.05 V.

The full cell was further tested for its rate performance, an important parameter for faster charging capabilities, as shown in **Figure 7**. The cycling rate performance is displayed in Figure 7a, in which the full cell was tested from 0.2 to 3.5 A $g^{-1}$, and then returned to 0.5 A $g^{-1}$ (the masses are related to the mass of Si only). The charge/discharge curves are shown in Figure S8 in the Supporting Information. The same rate testing is expressed in terms of energy density (Figure 7b). The specific capacity in Figure 7a was calculated in terms of Si mass in order to check how much of the capacity of Si tested in the half-cells is useful in the


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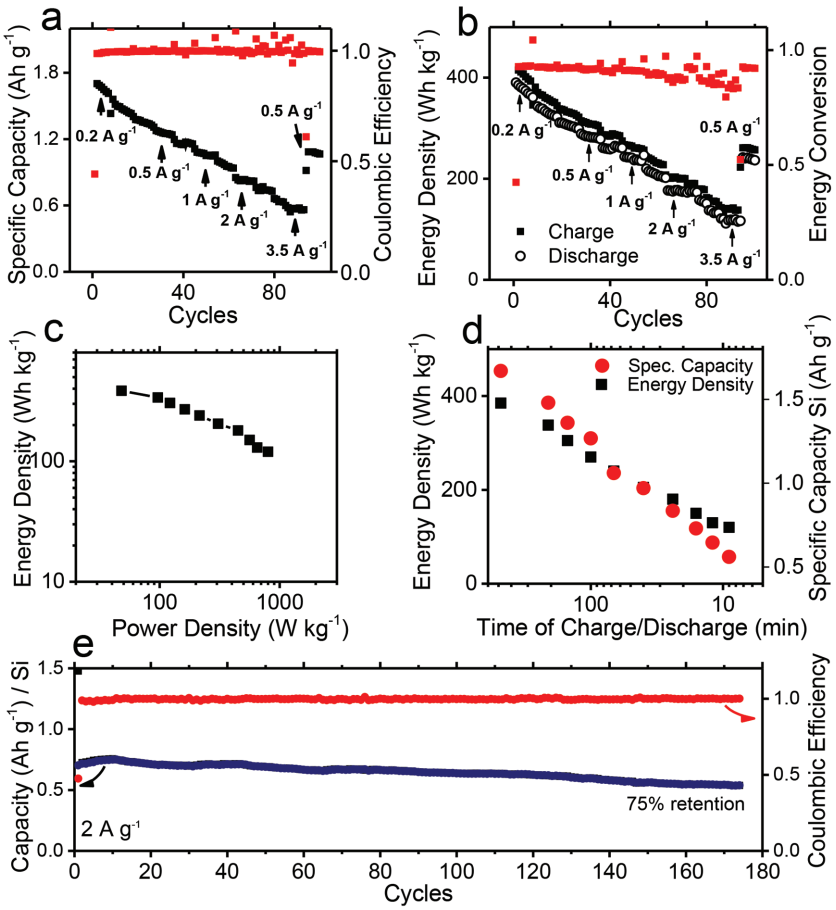



**Figure 7.** a,b) Cycling stability of a full battery, Si-NW/GNR anode and LCO-NW/GNR cathode, upon rate expressed in terms of specific capacity/coulombic efficiency and energy density/energy conversion, respectively. c) Ragone plot of the full battery. d) Energy density and specific capacity in terms of time of charge/discharge. Box indicates times of charge/discharge of less than 30 min. e) Stability of the full battery upon cycling at 2 A g$^{-1}$.

full cell configuration. At lower rates, 70% of the capacity is recovered in the full cell configuration. This proportion is high and is mainly determined by the upper limit of voltage (4.05 V), in order to prevent lithium plating over the Si electrode. The calculated energy density at lower rates (386 Wh kg$^{-1}$) is one the highest presented by anodes and is attributed here to the lower mass of Si in order to match the capacity of the cathode, and also to the similar voltage profile of Si, comparable to graphite.[14] Also, at lower rates, the coulombic efficiency is above 99.8% and the energy conversion is >86%, a significant and high value comparable to other battery technologies.[42] The coulombic efficiency measures the fraction of the amount of charge recovered after charging, while the energy conversion measures the amount of input energy (charge) converted to useful output energy (discharge). These values fluctuated at higher rates in the full cell, however they are restored when the full cell is retested at lower rates. The same graphs can be expressed in terms of time of charge and discharge (Figure 7d). Shorter times of charge/discharge of <1 h are still compatible with high capacity and high energy density (200 Wh kg$^{-1}$) when high rates are employed of >1 A g$^{-1}$. The stability in the full cell was tested after the rate testing in Figure 7e at a current density of 2 A g$^{-1}$, with a ≈75% retention observed after 174 cycles. The rate testing is also expressed in terms of a Ragone plot (Figure 7c), in which battery-compatible power densities are combined with high energy density, significantly higher when compared to commercial LIBs, which only show 100–250 Wh kg$^{-1}$.

## 3. Conclusion

We have designed a simple approach of filtration-made "papers" containing porous Si-NWs and GNRs. The homogenous conductive path of the papers is obtained from the nanowire–nanoribbon contact points. Compared with other highly complex assembly techniques to prepare Si anodes, this simple and scalable filtration-based method to prepared entangled wire-ribbon films forms a stable anode with superior capabilities in areal and volumetric capacity and mechanical flexibility. In addition, we observed that GNRs even at a high mass loading (≈20%) have a small contribution to the total volume of Si-NW/GNR paper, enabling a compact and conductive mat-like structure that can be reproduced over large thicknesses resulting in very high volumetric (≈4000 mA h cm$^{-3}$) and areal capacity (up to 11 mA h cm$^{-2}$). The same result could also be observed in LCO-NW papers with GNRs, and we demonstrated a high performance full battery with output voltage of ≈3.65 V. In order to have a high performance full battery, prelithiation methods were effective and necessary to compensate the low coulombic efficiency of the Si-NW/GNR anodes in the first cycle. The paper structure prepared from GNRs demonstrate that charges can percolate effectively through the void spaces of pure Si-NWs or LCO-NWs by GNRs entangling with the NWs, thus forming a permanent conductive path that is stable during the change in volume of the nanowires. The results presented here for interfacing 2D materials in hierarchical structures through simple graphene papers could also be applicable to other energy and sensing devices.

## 4. Experimental Section

*Silicon Nanowires Production*: Highly doped Si wafers (boron doped, resistivity <0.05 Ω cm) were used to generate Si-NW forests. The Si surface was cleaned using a bath sonicator (Cole Parmer Ultrasonic Cleaner, 12 W, 55 kHz) in soap solution (Contrex EZ, 1% mass in water) and then isopropanol (10 min each) and then dried using an air jet. The cleaned wafers were dipped into a 10% v/v hydrofluoric acid (HF) solution in water for 3 min and then directly dipped into the etching solution for 3 h. The etching solution was composed of 0.03 mol L$^{-1}$

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of $AgNO_3$ and 4.6 mol $L^{-1}$ of HF, adapted from the literature.[43,44] A total volume of 300 mL was used for etching an area of ≈20 $cm^2$. At the end of this period, the Si wafer had an Ag deposit covering the surface. The bulk top coat of Ag was removed by gently freeing the Ag layer using a spatula. The remaining Ag was removed by dipping the wafer in 1:1 water/$HNO_3$ 65% (total volume 100 mL) in water for 2–3 min to remove the remaining dendritic silver deposit and then flushing abundantly with distilled water. At this point the surface of the wafer was brownish black. The removal of the Si-NWs was done by sonicating the wafers in isopropanol (≈100 mL) using the same sonicator that was used for cleaning. The orange dispersion was filtered polyterafluoroethylene (PTFE, 0.45 μm) and vacuum dried to obtain a solid powder. After the Si-NW removal, the wafers were recycled by dipping into an aqueous solution of KOH 0.3 mol $L^{-1}$ for 3 h, washed liberally with water and then submitted to the same process.

*Silicon Nanowire/Graphene Nanoribbons Paper*: A dispersion of 0.1 mg $mL^{-1}$ of GNRs (EMD Merck, hydrogen-terminated edges) and 2 mg $mL^{-1}$ of Si-NWs in *N*-methyl-2-pyrrolidone (NMP) was prepared by sonication. The GNR dispersion was obtained using a tip sonicator (Mis Onix Sonicator 3000, tip radius 6.5 mm, 500 W, 20 kHz) and the Si-NW dispersion was obtained using a low power bath sonicator (Cole Parmer Ultrasonic Cleaner, 12 W, 55 kHz). A volume of 15 mL of the GNR dispersion was filtered (vacuum filtration on a 2 $cm^2$, 0.45 μm PTFE membrane filter) to prepare the bottom, pure GNR current collector. Then, a 1:2.5 (vol:vol) ratio of Si-NW and GNR dispersion were mixed and filtered over the GNR bottom layer, in order to prepare the top active layer of the electrode with 70–80 wt% of Si. The film was washed with methanol and ethanol to remove NMP residues and then the film was vacuum dried (<100 Torr) at room temperature. After the drying process, the film was peeled from the PTFE membrane and a volume of 50 μL of a 40 mg $mL^{-1}$ solution of poly(acrylic acid) (Sigma Aldrich, 35 wt% in water, $M_w$ 100 000) solution in methanol was added by dropping over the free-standing film which was then dried in room atmosphere. The film was further dried at 110 °C in vacuum (<100 Torr) for 6 h and directly transferred to a glove box.

*$LiCoO_2$ Nanowire Synthesis*: A mass of 1.87 g of $CoCl_2{\cdot}6H_2O$ and 0.48 g of urea ($CO(NH_2)_2$) were dissolved in 80 mL of distilled water. The solution was poured inside a Teflon lined hydrothermal reactor (steel autoclave) and the reaction was performed at 110 °C for 12 h. The resulting pink precipitate was filtered (0.45 μm PTFE membrane filter) and washed abundantly with water and then dried in an oven at 100 °C for 1 h. The powder was then annealed in air at 500 °C for 5 h for conversion to $Co_3O_4$ (black powder). The $Co_3O_4$ powder (0.58 g) and LiOH (191.5 g) in a molar proportion $Co_3O_4$:LiOH of 1:3.3 were then added to a flask with methanol (100 mL) and stirred for 3 h at room temperature. The solvent was evaporated and then the powder was submitted to a second annealing process in air by first heating to 450 °C for 3 h and then to 750 °C for another 3 h and then cooled to room temperature. The $LiCoO_2$-NW/GNR paper was prepared using the same method used for preparing Si anodes, with a mass ratio of 1:5 of GNR to $LiCoO_2$, according to the thickness of the film. The film was peeled from the PTFE membrane and a volume of a solution of poly(vinylidene fluoride) (PVDF, Sigma-Aldrich, $M_w$ 534 000) in NMP (10 mg $mL^{-1}$) was added to the film, in order that the PVDF composed 5% in mass of the total electrode film. Finally, the film was dried at 70 °C under vacuum for 12 h before use.

*Half-Cell and Full Cell Battery Assembly*: The electrochemical tests were performed in 2032 coin-type using Si-NW/GNR papers as the anode material and lithium foil as the counter and reference electrode. The area of the substrates was 0.5–1 $cm^2$. The electrolyte was composed of 1 mol $L^{-1}$ of $LiPF_6$ in ethylene carbonate:diethylene carbonate 1:1 and 5 wt% of fluoroethylene carbonate + 1 wt% lithium bis(oxalate)borate. The battery assembly was conducted inside a glove box with oxygen levels below 1 ppm and the separator was a Celgard membrane 2500. For sealing the coin cell, a pressure of 1000 psi was applied. For full batteries, the Si-NW/GNR anodes were prepared by two methods. In the first method, the anode was previously tested in half-cells for at least three cycles. The test was finished with the anode at the delithiated state. The battery was open inside the glove box and the anode removed and recombined with a LCO-NW/GNR cathode to compose the full battery. In the second method, the anode was prelithiated by adding a drop of electrolyte over the Si-NW/GNR and pressing (hand pressure) a Li foil on the top of the anode for 2 h. After 2 h, the Li foil was removed and the anode was combined with the LCO-NW/GNR cathode to produce the full battery. A 1:12 mass ratio of Si:$LiCoO_2$ was used. Both the cathode and anode had approximately the same geometric area (0.5–1 $cm^2$).

*Characterization*: The morphology and thickness measurements were performed using a scanning electron microscope (FEI, Quanta-FESEM 400) operated at 20 kV and a high-resolution TEM (JEOL JEM-2100F) operated at 200 kV. Raman spectra were acquired using a 514.5 nm excitation line with a Renishaw Raman microscope. The X-ray diffractograms were obtained using Rigaku D/Max Ultima II (Cu Kα radiation, 1.5418 Å). Current versus potential curves were measured at ambient conditions with a 4155C Agilent semiconductor parameter analyzer.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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